Transformada de derivadas

$$\mathscr{L}\{y'(t)\} = \int_0^{\infty} e^{-st} y'(t)dt = \lim_{b\to\infty} \int_0^{b} e^{-st} y'(t)dt$$

Resolviendo la integral de forma indefinida:

$u = e^{-st}$

$du = -se^{-st}dt$

$dv = y'(t)dt$

$v = \int y'(t)dt$

$v = y(t)$

$$\int e^{-st} y'(t)dt = y(t)e^{-st} - \int y(t)\left(-se^{-st}dt\right) + C$$

$$\int e^{-st} y'(t)dt = y(t)e^{-st} + s\int e^{-st} y(t)dt + C$$

$$\mathscr{L}\{y'(t)\} = \int_0^{\infty} e^{-st} y'(t)dt$$

$$= \lim_{b\to\infty} \int_0^{b} e^{-st} y'(t)dt$$

$$= \lim_{b\to\infty} \left[y(t)e^{-st} \right]\Big|_{t=0}^{b} + \lim_{b\to\infty} s\int_0^{b} e^{-st} y(t)dt$$

$$= \lim_{b\to\infty} \left[y(b)e^{-sb} - y(0)e^{0} \right] + s \lim_{b\to\infty} \int_0^{b} e^{-st} y(t)dt$$

$$\mathscr{L}\{y'(t)\} = 0 - y(0) + sY(s)$$

$$\boxed{\mathscr{L}\{y'(t)\} = sY(s) - y(0)}$$

$$\mathscr{L}\{y''(t)\} = ¿?$$

$$\mathscr{L}\{y''(t)\} = ¿?$$

$Sea\ g(t) = y'(t)$

$g'(t) = y''(t)$

$$\mathscr{L}\{y''(t)\} = \mathscr{L}\{g'(t)\} = sG(s) - g(0)$$

$$= s\mathscr{L}\{g(t)\} - g(0)$$

$$= s\mathscr{L}\{y'(t)\} - y'(0)$$

$$= s\left[sY(s) - y(0)\right] - y'(0)$$

$$\boxed{\mathscr{L}\{y''(t)\} = s^2Y(s) - sy(0) - y'(0)}$$

En general se tiene el siguiente teorema.

Teorema:

Si $f, f', f'', ..., f^{(n-1)}$ son continuas y de orden exponencial y si $f^{(n)}$ es continua por tramos en $[0,+\infty[$, entonces:

$$\mathscr{L}\left\{f^{(n)}(t)\right\} = s^n F(s) - s^{n-1} f(0) - s^{n-2} f'(0) - s^{n-3} f''(0) - ... - s f^{(n-2)}(0) - f^{(n-1)}(0)$$, donde $F(s) = \mathscr{L}\{f(t)\}$.

De otra forma:

$$\mathscr{L}\left\{y^{(n)}(t)\right\} = s^n Y(s) - s^{n-1} y(0) - s^{n-2} y'(0) - s^{n-3} y''(0) ... s y^{(n-2)}(0) - y^{(n-1)}(0)$$

Ejemplos:

$$\mathscr{L}\{y'''(t)\} = s^3 Y(s) - s^2 y(0) - s y'(0) - y''(0)$$

$$\mathscr{L}\left\{y^{(4)}(t)\right\} = s^4 Y(s) - s^3 y(0) - s^2 y'(0) - s y''(0) - y'''(0)$$

$$\mathscr{L}\left\{f^{(5)}(t)\right\} = s^5 F(s) - s^4 f(0) - s^3 f'(0) - s^2 f''(0) - s f'''(0) - f^{(4)}(0)$$

El exponente de" s " disminuye iniciando con un valor igual al orden de la ecuación diferencial

<u>Solución de Ecuaciones Diferenciales Lineales Utilizando la Transformada de Laplace</u>

Si se tiene que $a_n \frac{d^n y}{dt^n} + a_{n-1} \frac{d^{n-1} y}{dy^{n-1}} + ... + a_0 y = g(t)$ con

$y(0) = y_0,\ y'(0) = y_1,\ y''(0) = y_2, ...,\ y^{(n-1)}(0) = y_{n-1}$

Donde $a_i, i = 1,2,3,.....$ son constantes.

Para resolver este problema de **valor inicial**, se puede aplicar en ambos miembros de la ecuación diferencial la trasformada de Laplace, luego despejar $F(s)$ o lo que es lo mismo $Y(s)$. Por último aplicar la trasformada inversa de Laplace para encontrar $y(t)$ o $f(t)$.

Ejemplo

Resolver $y'' + y = \sqrt{2}sen(\sqrt{2}t)$ con $y(0) = 10 \quad y'(0) = 0$

Solución

$$y'' + y = \sqrt{2}sen(\sqrt{2}t)$$

Aplicando la transformada de Laplace a ambos lados

$$\mathscr{L}\{y'' + y\} = \mathscr{L}\left\{\sqrt{2}sen(\sqrt{2}t)\right\}$$

Por linealidad de la trasformada

$$\mathscr{L}\{y''\} + \mathcal{L}\{y\} = \sqrt{2}\,\mathscr{L}\left\{sen(\sqrt{2}t)\right\}$$

Aplicando el teorema de derivada

$$\left[\ s^2Y(s) - sy(0) - y'(0)\right] + Y(s) = \sqrt{2}\left(\frac{\sqrt{2}}{s^2 + \left(\sqrt{2}\right)^2}\right)$$

$$\left[\ s^2Y(s) - s(10) - 0\ \right] + Y(s) = \frac{2}{s^2 + 2}$$

$$s^2Y(s) - 10s + Y(s) = \frac{2}{s^2 + 2}$$

$$s^2Y(s) + Y(s) = \frac{2}{s^2 + 2} + 10s$$

$$Y(s)\left[s^2 + 1\right] = \frac{2 + 10s(s^2 + 2)}{s^2 + 2}$$

$$Y(s)\left[s^2 + 1\right] = \frac{2 + 10s^3 + 20s}{s^2 + 2}$$

$$Y(s) = \frac{2 + 20s + 10s^3}{(s^2 + 2)(s^2 + 1)}$$

Aplicando fracciones parciales obtenemos:

$$\frac{2+20s+10s^3}{(s^2+2)(s^2+1)} = \frac{As+B}{s^2+2} + \frac{(Cs+D)}{s^2+1}$$

$$2+20s+10s^3 = (As+B)(s^2+1)+(Cs+D)(s^2+2)$$

$$2+20s+10s^3 = As^3+Bs^2+As+B+Cs^3+Ds^2+2Cs+2D$$

$$2+20s+10s^3 = (A+C)s^3+(B+D)s^2+(A+2C)s+(B+2D)$$

$$\begin{cases} A+C=10 & \text{Ec(1)} \\ B+D=0 & \text{Ec(2)} \\ A+2C=20 & \text{Ec(3)} \\ B+2D=2 & \text{Ec(4)} \end{cases}$$

Ec(3)-Ec(1)

$$\begin{array}{r} A+2C=20 \\ \underline{-A-C=-10} \\ \boxed{C=10} \end{array}$$

Sustituyendo este valor de C en EC (1)

$$A+C=10$$
$$A+10=10$$
$$\boxed{A=0}$$

Ec(4)-Ec(2)

$$\begin{array}{r} B+2D=2 \\ \underline{-B-D=0} \\ \boxed{D=2} \end{array}$$

Sustituyendo este valor de D en EC (2)

$$B+D=0$$
$$B+2=0$$
$$\boxed{B=-2}$$

$$\frac{2+20s+10s^3}{(s^2+2)(s^2+1)} = \frac{(0)s-2}{s^2+2} + \frac{10s+2}{s^2+1}$$

$$Y(s) = -\frac{2}{s^2+2} + 10\frac{s}{s^2+1} + 2\frac{1}{s^2+1}$$

Ahora aplicamos la transformada inversa de Laplace

$$\mathscr{L}^{-1}\{F(s)\} = \mathscr{L}^{-1}\left\{-\frac{2}{s^2+2}+10\frac{s}{s^2+1}+2\frac{1}{s^2+1}\right\}$$

$$f(t) = \mathscr{L}^{-1}\left\{-\frac{2}{s^2+2}\right\}+10\mathscr{L}^{-1}\left\{\frac{s}{s^2+1}\right\}+2\mathscr{L}^{-1}\left\{\frac{1}{s^2+1}\right\}$$

$$f(t) = -\frac{2}{\sqrt{2}}\mathscr{L}^{-1}\left\{\frac{\sqrt{2}}{s^2+2}\right\}+10\mathscr{L}^{-1}\left\{\frac{s}{s^2+1}\right\}+2\mathscr{L}^{-1}\left\{\frac{1}{s^2+1}\right\}$$

La solución de la ecuación diferencial es $\boxed{y(t) = -\sqrt{2}sen(\sqrt{2}t)+10\cos(t)+2sen(t)}$

Ejemplo:

Resolver $2y'''+3y''-3y'-2y=e^{-t}$ con las condiciones iniciales $y(0)=0,\ y'(0)=0,\ y''(0)=1$

Solución

Aplicando la transformada de Laplace en ambos miembros de la ecuación diferencial:

$$\mathscr{L}\{2y'''+3y''-3y'-2y\} = \mathscr{L}\{e^{-t}\}$$

Por la linealidad de la trasformada tenemos

$$2\mathscr{L}\{y'''\}+3\mathscr{L}\{y''\}-3\mathscr{L}\{y'\}-2\mathscr{L}\{y\} = \mathscr{L}\{e^{-t}\}$$

Luego aplicando el teorema de la transformada de una derivada a los tres términos del lado izquierdo, obtenemos:

$$2\left[s^3Y(s)-s^2y(o)-sy'(0)-y''(0)\right]+3\left[s^2Y(s)-sy(0)-y'(0)\right]-3\left[sY(s)-y(0)\right]-2Y(s) = \frac{1}{s+1}$$

$$2\left[s^3Y(s)-s^2(o)-s(0)-1\right]+3\left[s^2Y(s)-s(0)-0\right]-3\left[sY(s)-0\right]-2Y(s) = \frac{1}{s+1}$$

$$2\left[s^3Y(s)-1\right]+3\left[s^2Y(s)\right]-3\left[sY(s)\right]-2Y(s) = \frac{1}{s+1}$$

$$2s^3Y(s)-2+3s^2Y(s)-3sY(s)-2Y(s) = \frac{1}{s+1}$$

Sacando factor común Y(s)

$$Y(s)\left(2s^3+3s^2-3s-2\right)=\frac{1}{s+1}+2$$

$$Y(s)\left(2s^3+3s^2-3s-2\right)=\frac{1+2s+2}{s+1}$$

$$Y(s)=\frac{3+2s}{(s+1)(2s^3+3s^2-3s-2)}$$

Utilizando división sintética.

$2s^3+3s^2-3s-2$

2	3	−3	−2	1
	2	5	2	
2	5	2	0	−2
	−4	−2		
2	1	0		

$$2s^3+3s^2-3s-2=(s-1)(s+2)(2s+1)$$

$$Y(s)=\frac{3+2s}{(s+1)(s+2)(s-1)(2s+1)}$$

Ahora aplicamos fracciones parciales y obtenemos:

$$\frac{3+2s}{(s+1)(s+2)(s-1)(2s+1)}=\frac{A}{s+1}+\frac{B}{s+2}+\frac{C}{s-1}+\frac{D}{2s+1}$$

$$3+2s=A(s+2)(s-1)(2s+1)+\mathrm{B}(s+1)(s-1)(2s+1)+\mathrm{C}(s+1)(s+2)((2s+1)+\mathrm{D}(s+1)(s+2)(s-1)$$

Si $s=-1,\ 3+2(-1)=A(-1+2)(-1-1)\left(2(-1)+1\right)$

$1=A(1)(-2)(-1)$

$$\boxed{A=\frac{1}{2}}$$

Si $s=-2,\ 3+2(-2)=\mathrm{B}(-2+1)(-2-1)\left(2(-2)+1\right)$

$-1=B(-1)(-3)(-3)$

$$\boxed{B=\frac{1}{9}}$$

Si $s=1,\ 3+2(1)=\mathrm{C}(1+1)(1+2)\left(2(1)+1\right)$

$5=C(2)(3)(3)$

$$\boxed{C=\frac{5}{18}}$$

Si $S=-\frac{1}{2},\ 3+2\left(-\frac{1}{2}\right)=\mathrm{D}\left(-\frac{1}{2}+1\right)\left(-\frac{1}{2}+2\right)\left(-\frac{1}{2}-1\right)$

$2=D\left(\frac{1}{2}\right)\left(\frac{3}{2}\right)\left(-\frac{3}{2}\right)$

$$\boxed{D=-\frac{16}{9}}$$

$$Y(s)=\frac{3+2s}{(s+1)(s+2)(s-1)(2s+1)}=\frac{\frac{1}{2}}{s+1}+\frac{\frac{1}{9}}{s+2}+\frac{\frac{5}{18}}{s-1}+\frac{-\frac{16}{9}}{2s+1}$$

Por último al aplicar la Trasformada inversa a la ecuación anterior.

$$\mathscr{L}^{-1}\{Y(s)\} = \mathscr{L}^{-1}\left\{\frac{3+2s}{(s+1)(s+2)(s-1)(2s+1)}\right\}$$

$$= \frac{1}{2}\mathscr{L}^{-1}\left\{\frac{1}{s+1}\right\} + \frac{1}{9}\mathscr{L}^{-1}\left\{\frac{1}{s+2}\right\} + \frac{5}{18}\mathscr{L}^{-1}\left\{\frac{1}{s-1}\right\} - \frac{16}{9}\mathscr{L}^{-1}\left\{\frac{1}{2s+1}\right\}$$

$$y(t) = \frac{1}{2}e^{-t} + \frac{1}{9}e^{-2t} + \frac{5}{18}e^{t} - \frac{16}{9}\mathscr{L}^{-1}\left\{\frac{1}{2\left(s+\frac{1}{2}\right)}\right\}$$

$$y(t) = \frac{1}{2}e^{-t} + \frac{1}{9}e^{-2t} + \frac{5}{18}e^{t} - \frac{16}{18}\mathscr{L}^{-1}\left\{\frac{1}{\left(s+\frac{1}{2}\right)}\right\}$$

La solución de la ecuación diferencial es $\boxed{y(t) = \frac{1}{2}e^{-t} + \frac{1}{9}e^{-2t} + \frac{5}{18}e^{t} - \frac{8}{9}e^{-\frac{1}{2}t}}$

Ejemplo: Resolver $y'' + 4y = 0, \quad y_{(\pi)} = 1, \quad y_{\left(\frac{\pi}{4}\right)} = -1$

Solución

$$\mathscr{L}\{y'' + 4y\} = \mathscr{L}\{0\}$$

$$s^2Y(s) - sy(0) - y'(0) + 4Y(s) = 0$$

$$\left(s^2+4\right)Y(s) - y(0)\,s - y'(0) = 0$$

$$\left(s^2+4\right)Y(s) - as - b = 0$$

$$\left(s^2+4\right)Y(s) = as + b$$

$$Y(s) = \frac{as+b}{s^2+4}$$

$$Y(s) = \frac{as}{s^2+4} + \frac{b}{s^2+4}$$

$$Y(s) = a\left(\frac{s}{s^2+4}\right) + b\left(\frac{1}{s^2+4}\right)$$

$$\mathscr{L}^{-1}\{Y(s)\}=a\mathscr{L}^{-1}\left\{\frac{s}{s^2+4}\right\}+b\mathscr{L}^{-1}\left\{\frac{1}{s^2+4}\right\}$$

$$y(t)=a\cos(2t)+\frac{b}{2}sen(2t)$$

$$y(\pi)=1\rightarrow 1=a(1)+\frac{b}{2}(0)\rightarrow \boxed{a=1}$$

$$y\left(\frac{\pi}{4}\right)=-1\rightarrow -1=a(0)+\frac{b}{2}(1)\rightarrow \frac{b}{2}=-1$$

$$\boxed{y(t)=\cos(2t)-sen(2t)}$$

Ejemplo: Resolver $y''+9y=\cos(t),\quad y_{(\pi)}=\frac{1}{8},\quad y'_{(\pi)}=-3$

Solución

$$\mathscr{L}\{y''+9y\}=\mathscr{L}\{\cos(t)\}$$

$$s^2Y(s)-sy(0)-y'(0)+9Y(s)=\frac{s}{s^2+1}$$

$$\left(s^2+9\right)Y(s)-\underset{a}{y(0)}\,s-\underset{b}{y'(0)}=\frac{s}{s^2+1}$$

$$\left(s^2+9\right)Y(s)-as-b=\frac{s}{s^2+1}$$

$$\left(s^2+9\right)Y(s)=\frac{s}{s^2+1}+as+b$$

$$Y(s)=\frac{s}{\left(s^2+1\right)\left(s^2+9\right)}+\frac{as+b}{s^2+9}$$

$$Y(s)=\frac{s}{\left(s^2+1\right)\left(s^2+9\right)}+a\left(\frac{s}{s^2+9}\right)+\frac{b}{3}\left(\frac{3}{s^2+9}\right)$$

Usando fracciones parciales en el término $\frac{s}{\left(s^2+1\right)\left(s^2+9\right)}$

$$\frac{s}{\left(s^2+1\right)\left(s^2+9\right)}=\frac{As+B}{s^2+1}+\frac{Cs+D}{s^2+9}$$

$$s=\left(As+B\right)\left(s^2+9\right)+\left(Cs+D\right)\left(s^2+1\right)$$

$$s=\left(A+C\right)s^3+\left(B+D\right)s^2+\left(9A+C\right)s+\left(9B+D\right)$$

$$A+C=0,\quad B+D=0,\quad 9A+C=1,\quad 9B+D=0$$

$$\boxed{A=\frac{1}{8},\quad B=0,\quad C=-\frac{1}{8},\quad D=0}$$

$$\frac{s}{\left(s^2+1\right)\left(s^2+9\right)}=\frac{1}{8}\left(\frac{s}{s^2+1}\right)-\frac{1}{8}\left(\frac{s}{s^2+9}\right)$$

$$Y(s)=\frac{s}{\left(s^2+1\right)\left(s^2+9\right)}+a\left(\frac{s}{s^2+9}\right)+\frac{b}{3}\left(\frac{3}{s^2+9}\right)$$

$$Y(s)=\frac{1}{8}\left(\frac{s}{s^2+1}\right)-\frac{1}{8}\left(\frac{s}{s^2+9}\right)+a\left(\frac{s}{s^2+9}\right)+\frac{b}{3}\left(\frac{3}{s^2+9}\right)$$

Aplicando la transformada inversa

$$Y(s)=\frac{1}{8}\left(\frac{s}{s^2+1}\right)-\frac{1}{8}\left(\frac{s}{s^2+9}\right)+a\left(\frac{s}{s^2+9}\right)+\frac{b}{3}\left(\frac{3}{s^2+9}\right)$$

$$y(t)=\frac{1}{8}\cos(t)-\frac{1}{8}\cos(3t)+a\cos(3t)+\frac{b}{3}sen(3t)$$

$$y(\pi)=\frac{1}{8}\rightarrow\frac{1}{8}=\frac{1}{8}(-1)-\frac{1}{8}(-1)+a(-1)+\frac{b}{3}(0)\rightarrow\boxed{a=-\frac{1}{8}}$$

$$y'(t)=-\frac{1}{8}sen(t)+\frac{3}{8}sen(3t)-3asen(3t)+b\cos(3t)$$

$$y'(\pi)=-3\rightarrow-3=-\frac{1}{8}(0)+\frac{3}{8}(0)-3a(0)+b(-1)\rightarrow b=3$$

$$y(t)=\frac{1}{8}\cos(t)-\frac{1}{8}\cos(3t)-\frac{1}{8}\cos(3t)+sen(3t)$$

$$\boxed{y(t)=\frac{1}{8}\cos(t)-\frac{1}{4}\cos(3t)+sen(3t)}$$

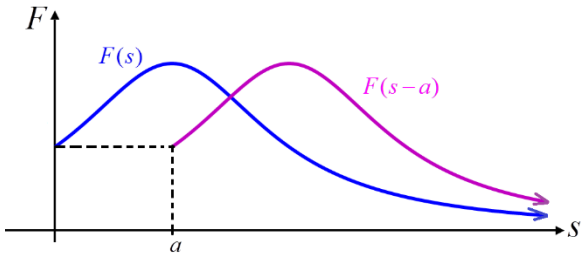
Teoremas de Traslación

Teorema: (**Primer Teorema de Traslación**)

Si $F(s)=\mathscr{L}\{f(t)\}$ y a es cualquier número real, entonces $\mathscr{L}\{e^{at}f(t)\}=F(s-a)$

Puede utilizarse la siguiente simbología

$$\mathscr{L}\{e^{at}f(t)\}=\mathscr{L}\{f(t)\}\big|_{s\to s-a}=F(s)\big|_{s\to s-a}$$



Comentario

$$\begin{aligned}\mathscr{L}\{e^{at}f(t)\}&=\int_0^{+\infty}e^{-st}e^{at}f(t)dt\\&=\int_0^{+\infty}e^{at-st}f(t)dt\\&=\int_0^{+\infty}e^{-(s-a)t}f(t)dt\\&=\int_0^{+\infty}e^{-zt}f(t)dt,\quad \text{donde}\ \ z=s-a\\&=F(z)\end{aligned}$$

$$\boxed{\mathscr{L}\{e^{at}f(t)\}=F(s-a)}$$

Ejemplo

Evaluar $\mathscr{L}\left\{e^{-2t}\cos(4t)\right\}$

Solución

$$\mathscr{L}\left\{e^{-2t}\cos(4t)\right\} = \mathscr{L}\left\{\cos(4t)\right\}\Big|_{s\to s+2}$$

$$= \frac{s}{s^2+16}\Big|_{s\to s+2}$$

$$F(s) = \frac{s+2}{(s+2)^2+16}$$

Ejemplo

Encontrar $\mathscr{L}\left\{5^t\, sen(3t)\right\}$

Solución

Para poder utilizar el teorema de traslación haremos la siguiente transformación

$$5^t = e^{\ln 5^t} = e^{t\ln 5} = e^{(\ln 5)t}$$

Luego

$$\mathscr{L}\left\{5^t\, sen(3t)\right\} = \mathscr{L}\left\{e^{(\ln 5)t} sen(3t)\right\}$$

$$= \mathscr{L}\left\{sen(3t)\right\}\Big|_{s\to s-\ln 5}$$

$$= \frac{3}{s^2+9}\Big|_{s\to s-\ln 5}$$

$$F(s) = \frac{3}{(s-\ln 5)^2+9}$$

Ejemplo

Hallar $\mathscr{L}\left\{e^{4t}t^2\right\}$

Solución

$$\mathscr{L}\left\{e^{4t}t^2\right\} = \mathscr{L}\left\{t^2\right\}\Big|_{s\to s-4}$$

$$= \frac{2!}{s^{2+1}}\Big|_{s\to s-4}$$

$$F(s) = \frac{2}{(s-4)^3}$$

<u>Forma Inversa del Primer Teorema de Traslación</u>

$$\mathscr{L}^{-1}\left\{F(s-a)\right\} = \mathscr{L}^{-1}\left\{F(s)\Big|_{s\to s-a}\right\} = e^{at}\mathscr{L}^{-1}\left\{F(s)\right\} = e^{at}f(t)$$

Ejemplo:

Hallar $\mathscr{L}^{-1}\left\{\dfrac{1}{(s+2)^3}\right\}$

Solución

$$\mathscr{L}^{-1}\left\{\frac{1}{(s+2)^3}\right\}=\mathscr{L}^{-1}\left\{\frac{1}{s^3}\Big|_{s\to s+2}\right\}=e^{-2t}\,\mathscr{L}^{-1}\left\{\frac{1}{s^3}\right\}=e^{-2t}\left(\frac{1}{2}\right)\mathscr{L}^{-1}\left\{\frac{2}{s^3}\right\}=\frac{1}{2}e^{-2t}t^2$$

$$\boxed{\mathscr{L}^{-1}\left\{\frac{1}{(s+2)^3}\right\}=\frac{1}{2}e^{-2t}t^2}$$

Ejemplo: Hallar $\mathscr{L}^{-1}\left\{\dfrac{s}{s^2+4s+5}\right\}$

Solución

Para poder utilizar la forma inversa anterior vamos a transformar la expresión del denominador, Haciendo una completación de trinomio cuadrado perfecto ya que no se puede factorizar porque es irreducible.

$$\frac{s}{s^2+4s+5}=\frac{s}{s^2+4s+4+1}=\frac{s}{(s+2)^2+1}$$

$$=\frac{s+2-2}{(s+2)^2+1}$$

$$=\frac{s+2}{(s+2)^2+1}+\frac{-2}{(s+2)^2+1}$$

$$\mathscr{L}^{-1}\left\{\frac{s}{s^2+4s+5}\right\}=\mathscr{L}^{-1}\left\{\frac{s+2}{(s+2)^2+1}\right\}+\mathscr{L}^{-1}\left\{\frac{-2}{(s+2)^2+1}\right\}$$

$$=\mathscr{L}^{-1}\left\{\frac{s}{s^2+1}\Big|_{s\to s+2}\right\}-2\,\mathscr{L}^{-1}\left\{\frac{1}{s^2+1}\Big|_{s\to s+2}\right\}$$

$$=e^{-2t}\,\mathscr{L}^{-1}\left\{\frac{s}{s^2+1}\right\}-2e^{-2t}\,\mathscr{L}^{-1}\left\{\frac{1}{s^2+1}\right\}$$

$$f(t)=e^{-2t}\cos(t)-2e^{-2t}sen(t)$$